

GANESH GUPTA

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Experience

Oracle

Jun 2025 – Present

Software Engineer, AI Platform

- Build agentic AI systems (LangGraph, CrewAI, AutoGen, OpenAI Agents SDK) that generate, validate, and deploy configuration changes autonomously across production infrastructure.
- Write infrastructure/systems-level code in Go, Rust, and C++ for large-scale cluster scheduling, resource allocation, and fault-tolerant placement logic under scarcity constraints across AWS, GCP, and CoreWeave.
- Built a deployment-gated evaluation framework enforcing guardrails and observability across autonomous system outputs, blocking 15% of failed releases before rollout.
- Own full software development lifecycle: coding standards, code reviews, CI/CD, and production operations, working in Agile sprints with Git-based version control.

QureAI (AI Startup)

Mar 2026 – Present

ML Infrastructure Engineer

- Built custom evaluators for RAG/agent workflows tracking drift, latency, and regression across model versions, catching failures before they reached production.
- Improved semantic retrieval recall 61% to 89% on a 2K-query eval set by rebuilding the chunking pipeline with hybrid dense-sparse retrieval and re-ranking.

Nomura Research Institute Financial Technologies

Sep 2019 – Aug 2023

Software Engineer

- Architected and operated 25+ production C++ and SQL-backed microservices for 1M+ users, sustaining 99.9% uptime over 4 years on AWS (EC2, RDS, SQS).
- Drove a redesign of nightly data aggregation, cutting processing from 8 minutes to 90 seconds via indexed views and batch partitioning.
- Standardized failure-handling contracts (retry, timeout, propagation semantics) across 25+ distributed services, reducing production escalations 60% in one quarter.
- Mentored engineers through code reviews and design write-ups, driving consistent engineering standards.

Projects

Distributed KV Store & VPC Control Plane Simulator | github.com/ganeshhgupta/distributed-kv-store

- Built a distributed key-value store in C++/Go implementing Raft consensus, linearizable reads, and log compaction; simulated VPC control-plane semantics for network partition and leader-election testing.

RootCause AI | github.com/ganeshhgupta/rootcause-ai

- Multi-agent system diagnosing production incidents via tool-use chains; routes alerts to specialized sub-agents and synthesizes evidence-backed root cause reports, reducing mean time to resolution.

LangFetch | tinyurl.com/LangFetch

- Agent compiling natural-language questions into dependency-ordered SQL subqueries with vector retrieval and hybrid re-ranking; TypeScript/Node.js backend, PostgreSQL.

Education

University of Texas at Arlington

Aug 2023 – May 2025

MS, Computer Science, GPA 4.0/4.0

MCKV Institute of Engineering

Jul 2016 – Jul 2020

BTech, Computer Science

Skills

Languages: C++, Java, Go, Python, SQL

Distributed Systems: Raft Consensus, Object-Oriented Design, Fault-Tolerant Scheduling, Multi-Threaded Systems

Agentic AI & GenAI: LangGraph, CrewAI, AutoGen, OpenAI Agents SDK, RAG, Model/Agent Evaluation

Cloud & Platform: AWS, GCP, CoreWeave, Docker, Kubernetes, CI/CD

Certifications

AWS Certified Developer Associate | GCP Professional MLOps Engineer